



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

The world's smallest addressable LED

The SMD LED - Addressable RGB APA-104-1010 measuring 1.0mm x 1.0mm is a development in the trend of miniaturisation. Despite its size, it maintains the same functional-



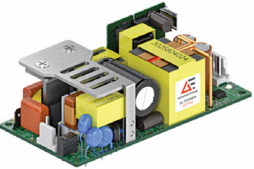
ity as its larger counterparts, making it an option for applications where space is critical. These LEDs are suitable for projects involving wearable technology, drones, or any device where space and weight are limited. Designers can now incorporate RGB lighting without compromising the size or weight of their creations. The size allows for versatile PCB layouts, enabling new designs, especially in devices like drones and wearables. Additionally, the energy efficiency of these LEDs, with a maximum current draw of 15mA per LED, makes them suitable for battery-powered applications where conserving power is important.

Sparkfun

<https://www.sparkfun.com>

World's first off-the-shelf power supply

Advanced Energy has launched the NCF150 series of AC-DC power supplies. These units are designed to help medical equipment manufacturers meet the cardiac floating (CF)



requirements. The power supplies enable the design of medical devices that require direct cardiac contact, such as dialysis machines and electrosurgery

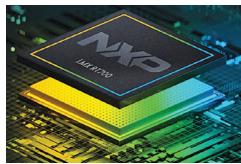
platforms, without custom solutions. The series includes features necessary for CF certification—high isolation voltage and low leakage current. They accept an input range of 85V AC to 264V AC and operate with over 90 percent full-load efficiency.

Advanced Energy

<https://ir.advancedenergy.com>

Two hundred times Edge AI performance multicore MCU

NXP has announced the i.MX RT700, a multicore MCU with an integrated eIQ neutron neural processing unit (NPU). This new series enhances the capabilities of its predecessors by lowering power consumption and increasing performance due to additional cores and architectural improvements. It features up to 7.5MB internal SRAM divided into 30 partitions for multicore access.



The series reduces power consumption by up to 70% compared to previous generations and supports the 1.2V eUSB standard for interaction with USB 2.0 devices. The i.MX RT700 targets applications that require sensing, DSP, and computing capabilities while maintaining low power consumption for extended battery life.

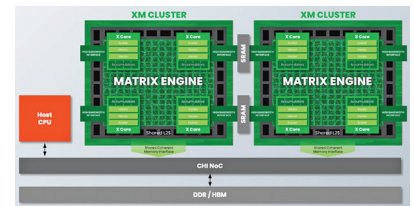
NXP Semiconductors

<https://www.nxp.com>

Semiconductor chips for enhanced AI workloads

SiFive, Inc. has launched the XM series, designed to enhance AI workloads.

This series is the first IP from SiFive to include a scalable AI matrix engine, aimed at reducing time to market for semiconductor companies developing system-on-chip solutions across sectors such as edge IoT, consumer devices,



electric and autonomous vehicles, data centres, and more. The XM series features a scalable AI compute engine that integrates scalar, vector, and matrix engines to optimize memory bandwidth usage. The series continues SiFive's standard of performance per watt, with each cluster containing four X-Cores capable of delivering 16 TOPS (INT8) or 8 TFLOPS (BF16) per GHz and providing 1TB/s of sustained memory bandwidth.

SiFive

<https://www.sifive.com>

The first satellite IoT-NTN cellular module

u-blox has introduced the SARA-S528NM10, a module that complies with both terrestrial network (TN) and non-terrestrial network (NTN) standards. This module provides global coverage with positioning that is



suitable for continuous or cyclic asset tracking and monitoring. The technology is also applicable for various IoT applications, including aftermarket telematics, industrial monitoring and